Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/26/2012
PI Name:	Kamotani, Yasuhiro Ph.D.		
Project Title:	Investigation into the Quantification and Control of Steady and Unsteady Flow Phenomena in Passive Gas-Liquid Cyclonic Separators		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
Flight Assignment:	NOTE: End date is now 5/31/2014, per J. McQuillen/GRC (Ed., 3/7/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan (Case Western Reserve University)		
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	Phase separators are devices designed to divide flows of intermingling gas and liquid into separate streams that are individually purely liquid or purely gas. Separation of the respective phases is of considerable interest to NASA as many spacecraft systems can only function effectively if the flow they receive is composed exclusively of the one phase that they are designed to accommodate. While separation occurs naturally on the earth, the absence of buoyancy in a microgravity environment allows the undesired mixtures of gas and liquid to remain combined unopposed. Due to this reluctance of the phases to separate, installation of some type of separation device becomes supremely important for many of the fluid systems that are used by NASA. Cyclonic separators provide the gas-liquid separatory action by swirling the multiphase flow – causing the gas to accumulate along the axis of the vortex as the denser liquid is forced to the walls – thereby allowing segregated extraction of the respective phases. Passive cyclonic separators utilize only the		

Task Description:	inertia of the incoming flow to accomplish this task and in doing so can avoid the moving parts and reliability concerns of their active brethren. While devices of this type have been successfully employed in experimental settings, more complete investigations and analyses are required before these devices can ever be seriously considered for use in operational space hardware. The objective of this study is to facilitate the maturation of passive cyclonic separation technology through exhaustive research of a newly designed gas-liquid phase separator evolved from NASA Glenn Research Center's (GRC) existing Cascade Cyclonic Separation Device (CSD-C). Combined experimental and numerical analyses will seek to quantitatively assess and delimit the range of operability of the new separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct – features that have yet to be completely explored in any such device. The approach is first to assess the range of operability of the new separator design through the careful construction of detailed flow separation maps using data gathered from reduced-gravity flight experimentation in conjunction with ancillary computational studies. Second, the transient and stability behavior of the device will be observed and analyzed. Special attention will be paid both to device-level conduct and also to behaviors that could manifest dysfunction elsewhere at the complete system level. A novel extension of the phase-field computational multiphase fluid dynamics (CMFD) method has been coupled to 1-D system-level multiphase solvers in order to permit discernment of these flow features. Lastly, with the ensuing understanding of the causes of any performance-affecting responses, new ways to control transient and unstable operations will be sought; in particular, the insight acquired should facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CMFD as a reliable design tool to further assist in the creation and analysis of new designs. The combination of the research activity included in this grant with the complementary activity of Dynaflow Inc. of Jessup, MD (Grant NNX09AI35G) has evolved into the Two-Phase Flow Separator Experiment (TPFSE). TPFSE represents a comprehensive investigation of passive cyclone separation technology – an investigation made possible by capabilities that only NASA and the International Space Station can offer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Phase separation has been identified as a critical mission-enabling capability for spacecraft systems ranging from power conversion to Environmental Control and Life Support Systems. As these systems have little tolerance for failure, extensive investigations and analyses are required before these devices can ever be seriously considered for employment in operational space hardware. Due to the advantages associated with passive vortex devices, cyclonic separators are also finding increasing use in terrestrial applications, such as in the petroleum industry where separators of this type are considered to be an emerging technology. The detailed study of the features of the complicated vortex-driven flow along with the modeling techniques under development for the microgravity investigations could equally be exploited for the advancement of these more earthly applications.
Task Progress:	Separator Design The project plan initially concerned experimentation and analysis with respect to the existing GRC CSD-C separator, but a design evolution, the Case Western Reserve University Separator (hereafter referred to as CWRU-SEP), has been developed instead. Continued studies of the CWRU-SEP separator have been performed in the past year and have shown that it is able to operate over a greater range of incoming flow rates and volume fractions than was possible with previous experimental designs. The improvements of CWRU-SEP arose by capitalizing on the results of our previous investigations into passive-separation technology. Continued experimentation with various other vortex separation designs has also been conducted throughout the past year of this grant. The work has consisted of ground-based experiments operating in a high flow rate, gravity-independent mode. A simple control volume analysis has been created to quantitatively assess and delimit the dynamic response of the separator. The basic fluid phenomena of the separator were studied for steady and transient conditions. New instrumentation including an Electrical Resistance Tomography system has been installed; experimental results thereof have compared well to the computational and analytical results. This work is providing additional guidance regarding subsequent 0-g operation. Beyond ground-based studies, reduced gravity experimentation will give further important information about separation of two phase flow. Toward that end, work on refurbishing the Two-Phase Flow Experiment (TPFE) apparatus at NASA GRC has been performed. With the separator installed inside it, the rig is now operating with new capabilities and instrumentation and is ready for a reduced-gravity flight campaign that has been scheduled for May 2012. Numerical and Analytical Modeling The CMFD studies of the fluid physics within the separation device conducted during the previous years of the grant have also been continued. The CWRU-SEP design optimization studies utilizing multifluid treatments have continued, but more advanced techniques have been created and refined to improve predictive capabilities. In particular, the custom phase-field surface implementation used previously has been updated and has been used to simulate some of the capillary instabilities that affect the functionality of the separator – features that could not be captured effectively with the previous modeling efforts. System-level conduct has been found to be crucial to the proper functionality of the separator – as such, updated efforts have been made to model the upstream and downstream components that are invariably present. To do so, one-dimensional two-phase flow modeling techniques have been coupled to the device-level three-dimensional simulations. Compressible HEM and multifluid approaches have both been employed. A simplified control-volume-based analytical approach has also been adopted to allow the prediction of the separator behavior. Using simple phenomenological descriptions, the analytical approach has provided useful scaling parameters and insight into device design. Satisfactory agreement with computational and experimental results has been found, and

	the nature of the transient flow features (including adjustment time scales) has been detailed. TPFSE This research activity has evolved into the Two-Phase Flow Separator Experiment (TPFSE). In accordance with the elevated status, the definition of the complete experimental requirements has been continued with respect to the objective of the emplacement of the experimental apparatus aboard the International Space Station. The changes to the separator design that necessarily accompany ISS installation are still being examined. Collaboration with NASA and contract personnel has furthermore continued in earnest to start defining the basic make-up of the complete system that is to be eventually employed on station.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
Articles in Peer-reviewed Journals	Hoyt N, Kang M, Kharraz A, Kamotani Y, Kadambi J. "Cyclonic two-phase flow separator experimentation and simulation for use in a microgravity environment." Journal of Physics: Conference Series 2011;327:012056. http://dx.doi.org/10.1088/1742-6596/327/1/012056 , Dec-2011
Papers from Meeting Proceedings	Hoyt N, Kang MF, Kharraz A, Kadambi J, Kamotani Y. "Dynamic Response of Passive Cyclonic Separator in Microgravity." Presented at the 50th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, Nashville, Tennessee, Jan. 9-12, 2012. 50th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, Nashville, Tennessee, Jan. 9-12, 2012. Paper AIAA 2012-0590. http://dx.doi.org/10.2514/6.2012-590 , Jan-2012